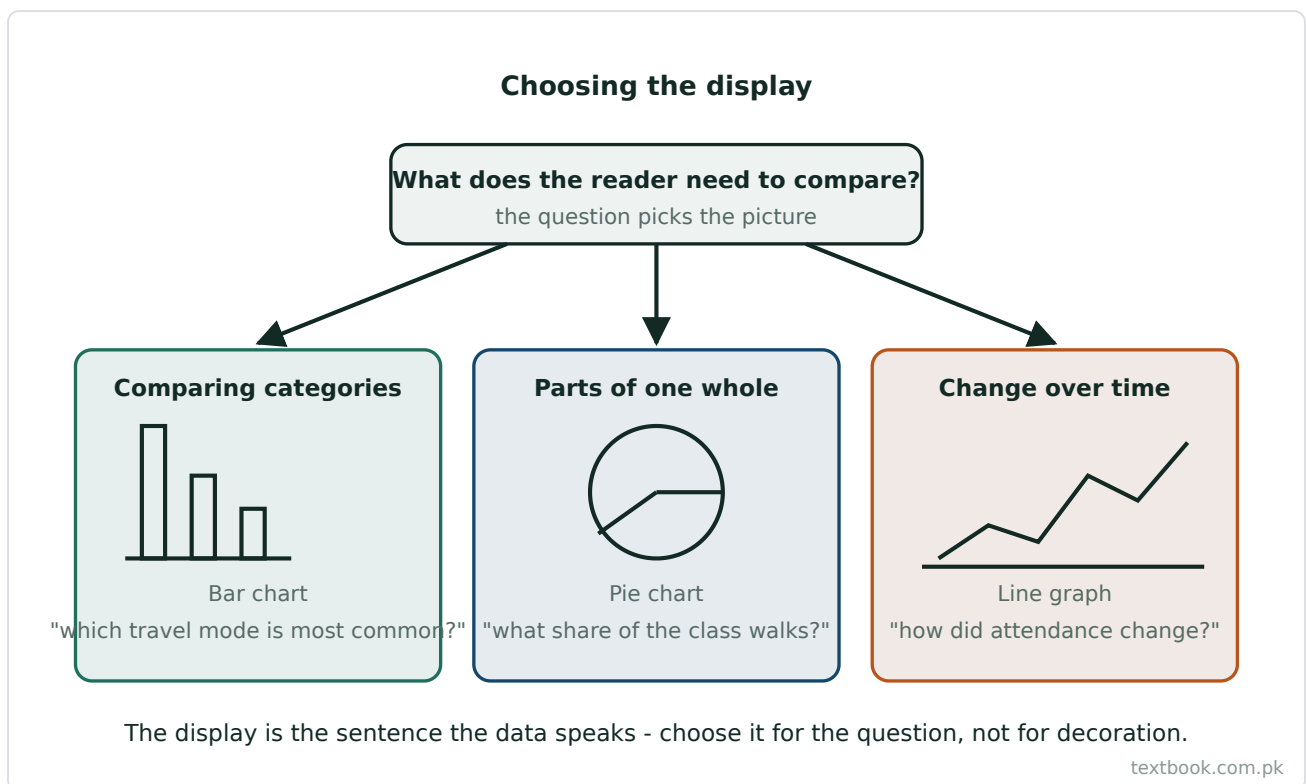


Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Topic 5.2 - Tables, graphs and charts

A real classroom situation



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Two pupils are asked to show the class's travel data to the Parent Teacher Association. One draws a pie chart of travel modes; the other draws a line graph of the same numbers. Both are neatly drawn. Both are wrong - not in their arithmetic, but in their choice. A pie chart answers "what share of the class walks?", and a line graph answers "how does something change over time?" Neither answers the question the PTA actually asks: "which mode is most common, and by how much?" - which is a bar chart's job. The display is not

decoration after the data; it is the sentence the data speaks. Choosing it is a quantitative decision, and this topic teaches the choice.

Explanation

Tables, graphs, and charts

Display	What it shows best	One pitfall
Frequency table	exact counts, nothing hidden	hard to see patterns at a glance
Bar chart	comparing categories side by side	truncated axis exaggerates gaps
Pie chart	shares of one whole	meaningless if not one whole
Line graph	change over time	implies continuity that may not exist
Pictograph	counts as icons, friendly to younger readers	every icon misleads

Reading a display means reading its construction, not just its impression.

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The table above is the topic in one view: five displays, what each shows best, and the pitfall each carries. Three of them deserve a closer look.

The bar chart. A bar chart compares categories with bars whose heights are the values. The class travel data (walk 16, van 11, rickshaw 8, bicycle 4, other 1) becomes five bars, and the eye answers "which is most common" before any number is read. The bar chart's one rule: bars start at zero. Start the vertical axis at 10 instead, and the walk bar towers over the van bar while the values differ by only 5 - the picture lies.

The pie chart. A pie chart shows parts of one whole as slices of a circle, with each slice's angle equal to its share of 360 degrees. The travel data at 40 pupils: walk is $16/40 = 40$ percent, so its slice is 40 percent of $360 = 144$ degrees. A

pie chart is honest only when the slices are shares of one meaningful whole - the class, the budget, the day. Slicing unrelated quantities into a pie is the chart's commonest abuse.

The line graph. A line graph tracks change over time, point by point, joined so the trend is visible. Attendance by day, marks by term, temperature by month: whenever the horizontal axis is time, the line graph is the default. The pictograph uses repeated icons (each stands for a fixed count) and suits young pupils and public notices, provided its key states exactly what one icon means.

A common misconception to correct. Many trainees believe "a graph's impression is the data". The impression is a product of choices: truncated axes, unequal bar widths, slices rearranged to hide a small one. Reading a display means reading its construction too - where does the axis start, what does one icon mean, what is left out. The PISA framework makes this exact demand: interpreting mathematics in context includes recognising how a display frames what it shows (OECD, n.d.).

Activity: Display the same data three ways

Work in pairs. Your tutor gives each pair the class travel data (walk 16, van 11, rickshaw 8, bicycle 4, other 1) and blank grids. Each pair draws (a) a bar chart with the axis starting at zero, (b) a pie chart with the slice angles computed, and (c) a pictograph with a stated key. Then the pairs swap drawings and answer one question about each: which single question does this display answer best? About 25 minutes.

Check your understanding

1. Name the display that best answers each question: (a) which travel mode is most common; (b) what share of the class walks; (c) how attendance changed across the week. (*Apply*)
2. Compute the pie-slice angle for a category that is 30 percent of the whole. (*Apply*)

3. A bar chart's vertical axis starts at 10 instead of 0. State in one sentence what the picture now exaggerates. (*Understand*)
4. A pictograph shows 3 bus icons for Class A and 5 for Class B, with the key "one icon = 10 pupils". State the two class counts, and one way the display could mislead if the key were missing. (*Analyze*)
5. Sketch (in words) the line graph for attendance of 92, 95, 88, 96, 91 percent over five days, and state what the line's dips and rises show. (*Apply*)

Summary

Each display answers one kind of question: bar charts compare categories, pie charts show shares of one whole, line graphs track change over time, pictographs carry a stated key into young readers' hands, and the frequency table hides nothing. The display is a choice with rules - bars start at zero, slices are shares of one whole, keys are stated - and reading a display means reading its construction, not just its impression.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can choose the right display for a given question and say why.
- ☐ I can draw a bar chart from a frequency table, and compute pie-slice angles.
- ☐ I can spot a misleading construction (truncated axis, missing key) and state what it hides.

Try this at your practicum school

Find one chart or table displayed in the school (a results board, a duty roster, a canteen price list). Identify which question it answers best, and one construction choice (axis start, ordering, key) that shapes the reader's impression. Bring a photo or sketch with your note.

Summative task

The school's five class strengths are 38, 42, 35, 40 and 45. The headteacher wants the managing committee to see which class is largest and by how much. In a short written answer: (a) name the best display for this question and draw or describe it precisely, (b) name one display that would be a poor choice and say what it would hide or exaggerate, and (c) compute the pie-slice angle for the largest class's share of total pupils, showing the working.

Mini-rubric (each criterion 0-2 marks):

- *Best display*: the bar chart chosen with a precise description (labels, axis from zero).
- *Poor choice analysed*: for example, a pie chart hides "by how much" between adjacent classes, or a pictograph's icons blur small differences - stated with a reason. **(Analyze)**
- *Slice angle*: total = 200; largest class $45/200 = 22.5$ percent; angle = $22.5\% \times 360 = 81$ degrees, with the working shown.

Further reading

- OECD. PISA mathematics framework. <https://www.oecd.org/pisa/>
- Pakistan Bureau of Statistics. (2023). *Provincial Census Report 2023 - Sindh*. Government of Pakistan. <https://www.pbs.gov.pk/> (report verified 2026-09-23)

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